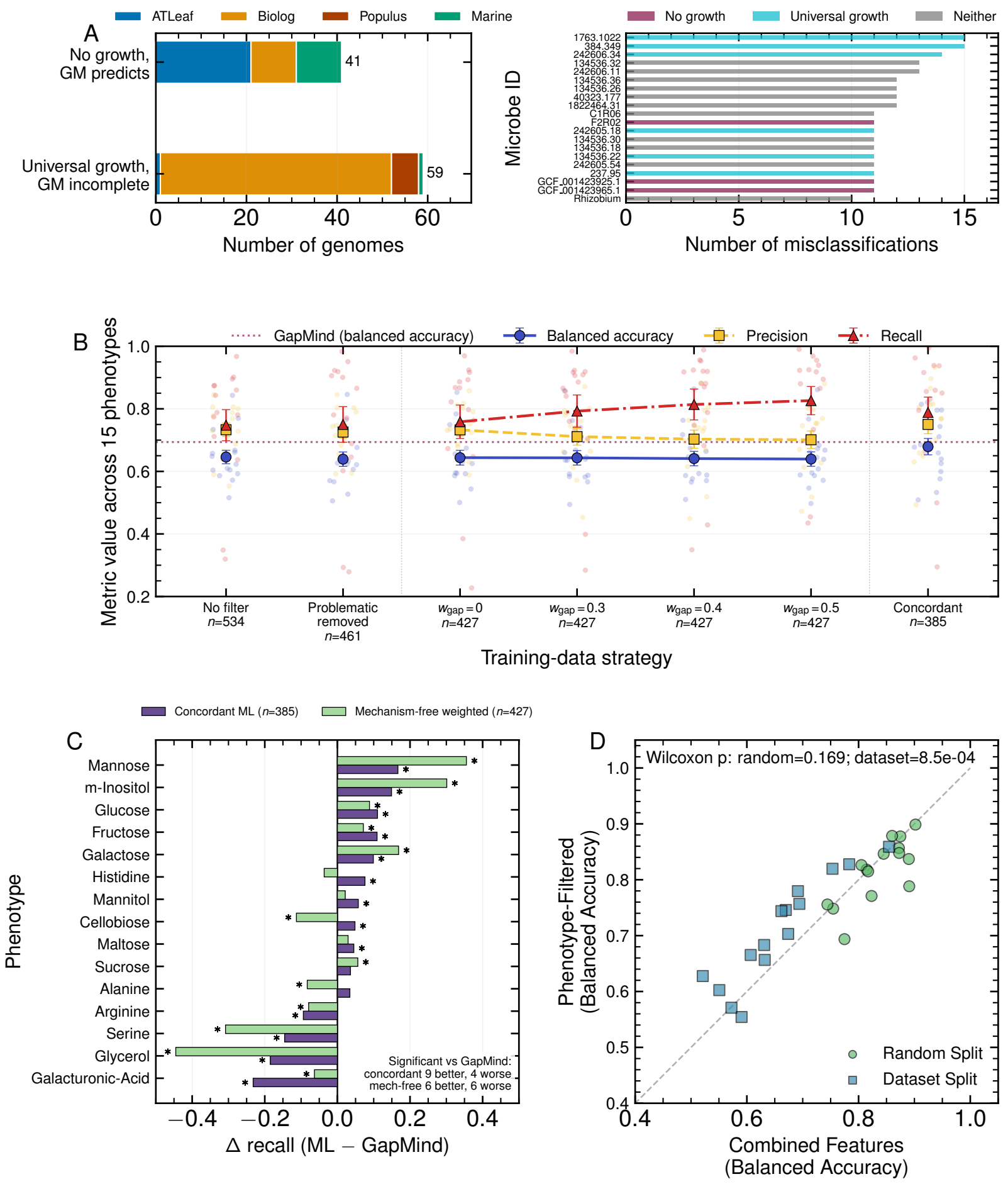

A
ATLeaf
Biolog
Populus
Marine
No growth, GM predicts
Universal growth, GM incomplete
41
59
0
20
40
60
Number of genomes
No growth
Universal growth
Neither
Microbe ID
1763.1022
384.349
242606.34
134536.32
242606.11
134536.36
134536.26
40323.177
1822464.31
C1R06
F2R02
242605.18
134536.30
134536.18
134536.22
242605.54
237.95
GCF_001423925.1
GCF_001423965.1
Rhizobium
0
5
10
15
Number of misclassifications
B
GapMind (balanced accuracy)
Balanced accuracy
Precision
Recall
Metric value across 15 phenotypes
1.0
0.8
0.6
0.4
0.2
No filter n=534
Problematic removed n=461
wgap = 0 n=427
wgap = 0.3 n=427
wgap = 0.4 n=427
wgap = 0.5 n=427
Concordant n=385
Training-data strategy
Concordant ML (n=385)
Mechanism-free weighted (n=427)
C
Mannose
m-Inositol
Glucose
Fructose
Galactose
Histidine
Mannitol
Cellobiose
Maltose
Sucrose
Alanine
Arginine
Serine
Glycerol
Galacturonic-Acid
Phenotype
Significant vs GapMind:
concordant 9 better, 4 worse
mech-free 6 better, 6 worse
−0.4
−0.2
0.0
0.2
0.4
Δ recall (ML − GapMind)
D
Wilcoxon p: random=0.169; dataset=8.5e-04
Phenotype-Filtered (Balanced Accuracy)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Random Split
Dataset Split
0.4
0.6
0.8
1.0
Combined Features (Balanced Accuracy)